Knowledge Representation

Stefan Schlobach
With slides from
Frank van Harmelen

Who am I?

- Stefan Schlobach
- Studied CS in Saarbrücken and Paris
- PhD in Logic from University of London, King’s College

- Associate Prof at VU, co-leading a group of ±15 researchers
- Knowledge Representation, Semantic Web, Knowledge Graphs, Knowledge Engineering

Examples of intelligent behaviour?

carry out complex reasoning
(solve physics problems,
prove theorems)

“but we do not mean that people can recognize familiar objects, or execute motor skills, or navigate around space, abilities we share with dogs and cats (and fish)”

conversations)

solving novel complex problems
(generating plans,
designing artifacts)

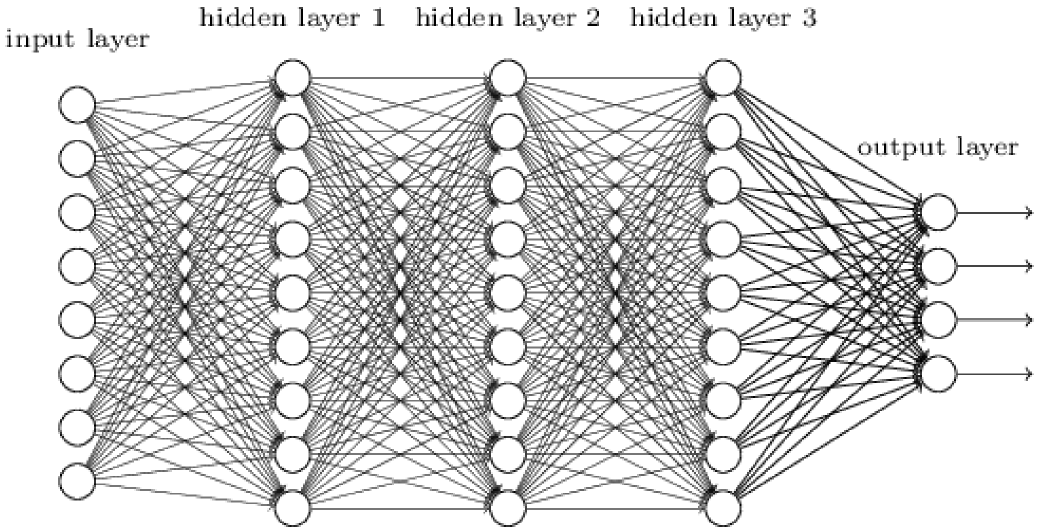
But isn’t modern AI all about Machine Learning ?

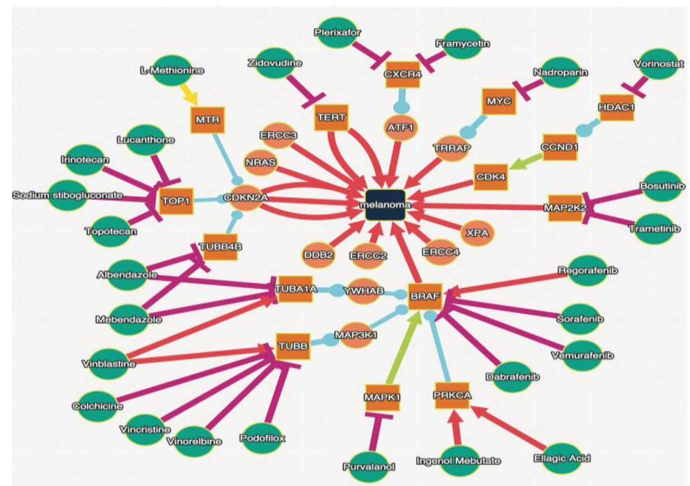
Two main lines of development in AI

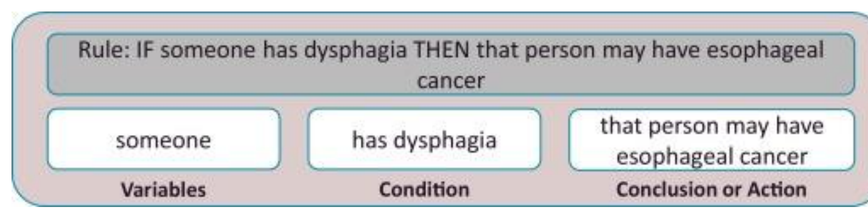
- symbolic representations
- statistical representation

There have been alternating cycles of one dominating over the other in different decades of the history of AI.

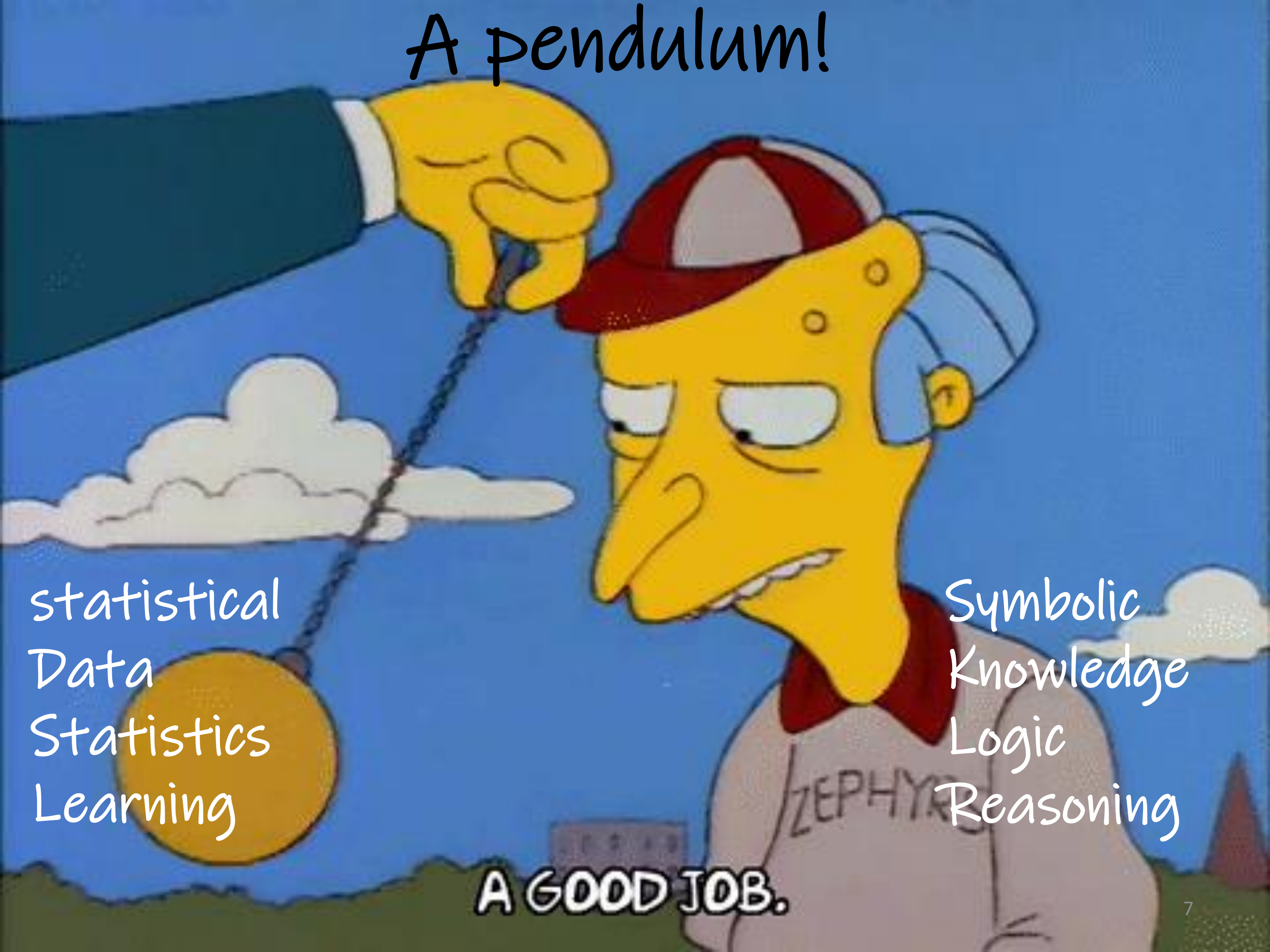
Statistical vs. Symbolic








A stairway?
statistical
Data
Statistics
Learning
Symbolic
Knowledge
Logic
Reasoning


A pendulum!
statistical
Data
Statistics
Learning
Symbolic
Knowledge
Logic
Reasoning
ZEPHYRS
A GOOD JOB.

Statistical vs. symbolic AI: very different types of applications

statistical:

- pattern recognition (images, sound, shapes)
- motor skills (robots)
- speech generation (sound)
- search engines

symbolic:

- planning (autonomous space missions)
- reasoning (diagnosis, design, decision support)
- language generation (conversations beyond Q&A)
- search engines

Audience participation…

Who was his predecessor?

KR

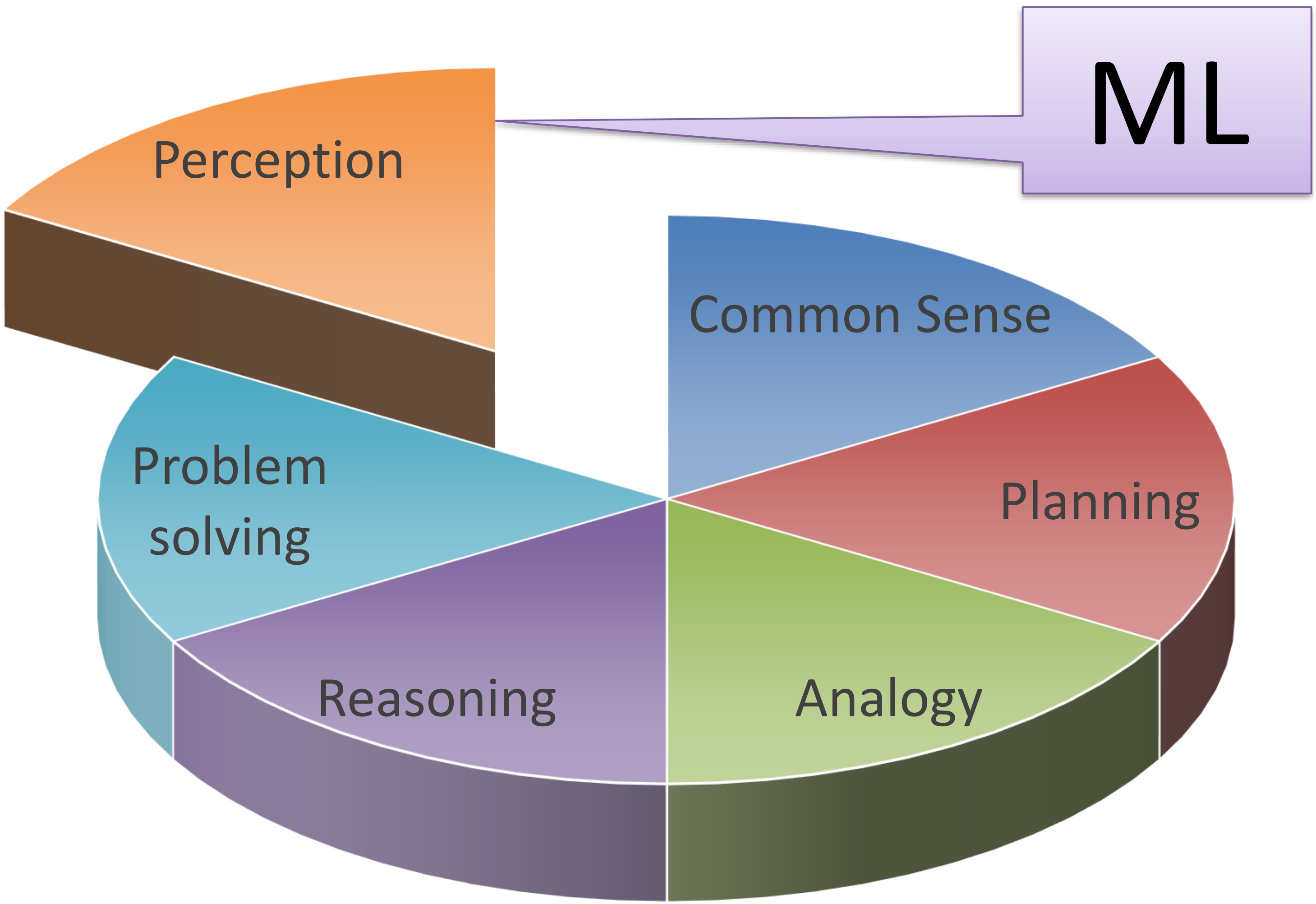
Who was his predecessor's predecessor?

Thinking slow

Thinking fast

ML

Human intelligence = thinking fast & thinking slow



Thinking fast
without thinking slow

Class: 793

Label: n04209133 **(shower cap)**

Certainty: 99.7%

& Weaknesses

	Symbolic	**statistical**
Construction	Human effort	Data hunger
Scaleable	+/-	+/-
Explainable	+	-
Generalisable	Performance cliff	Performance cliff

Strengths & Weaknesses

	Symbolic	statistical
Construction	**Human effort**	Data hunger
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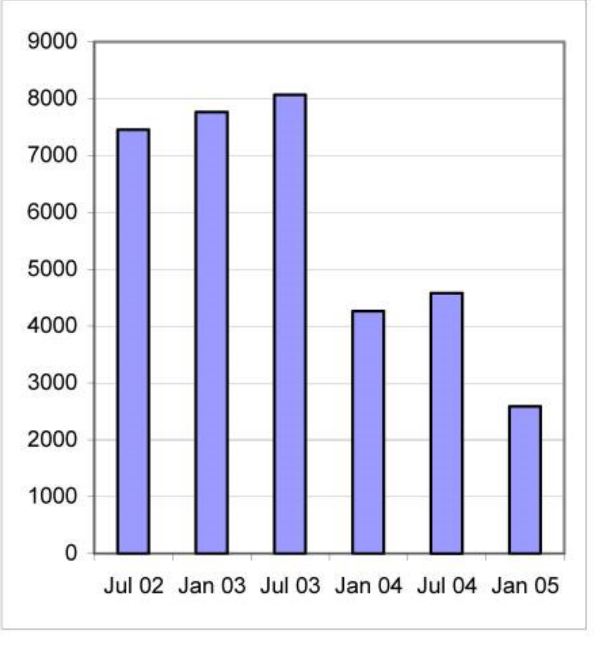
SNOMED CT
The global language of healthcare

300.00 definitions
40 years of effort,
10.000 updates every years



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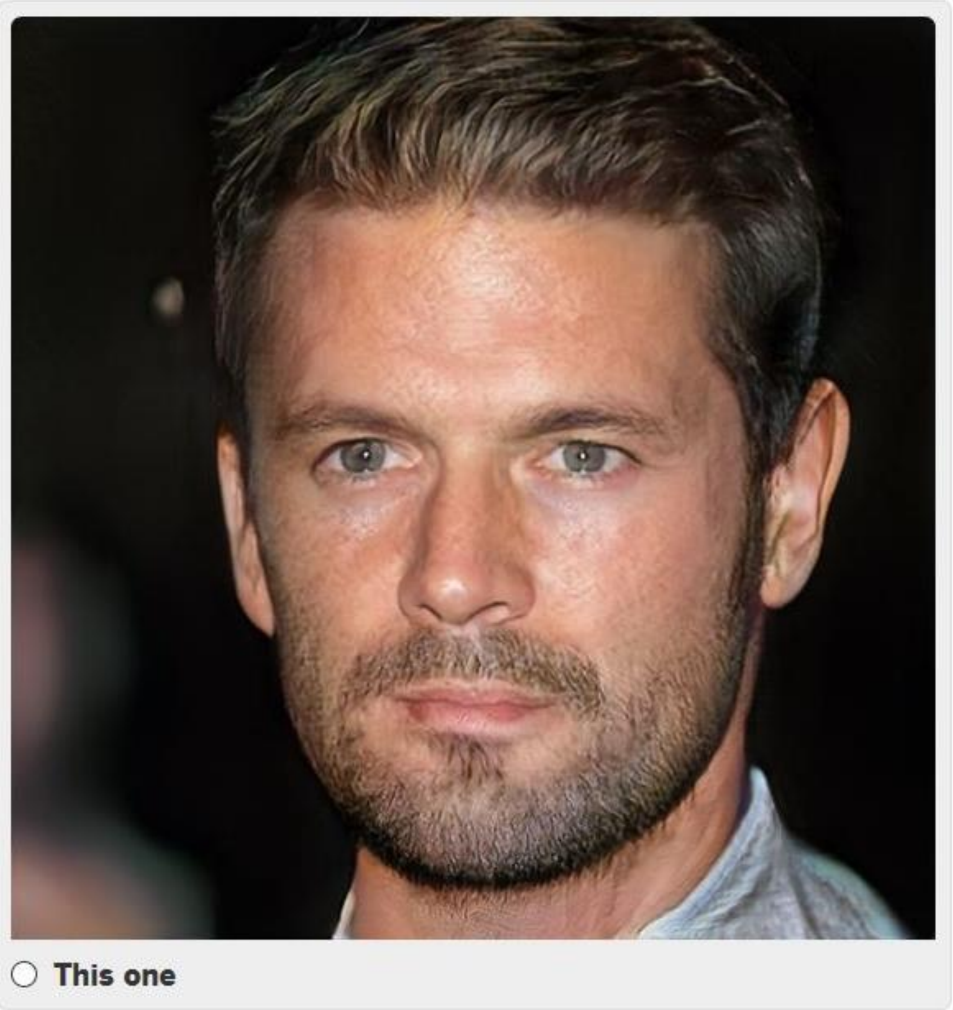

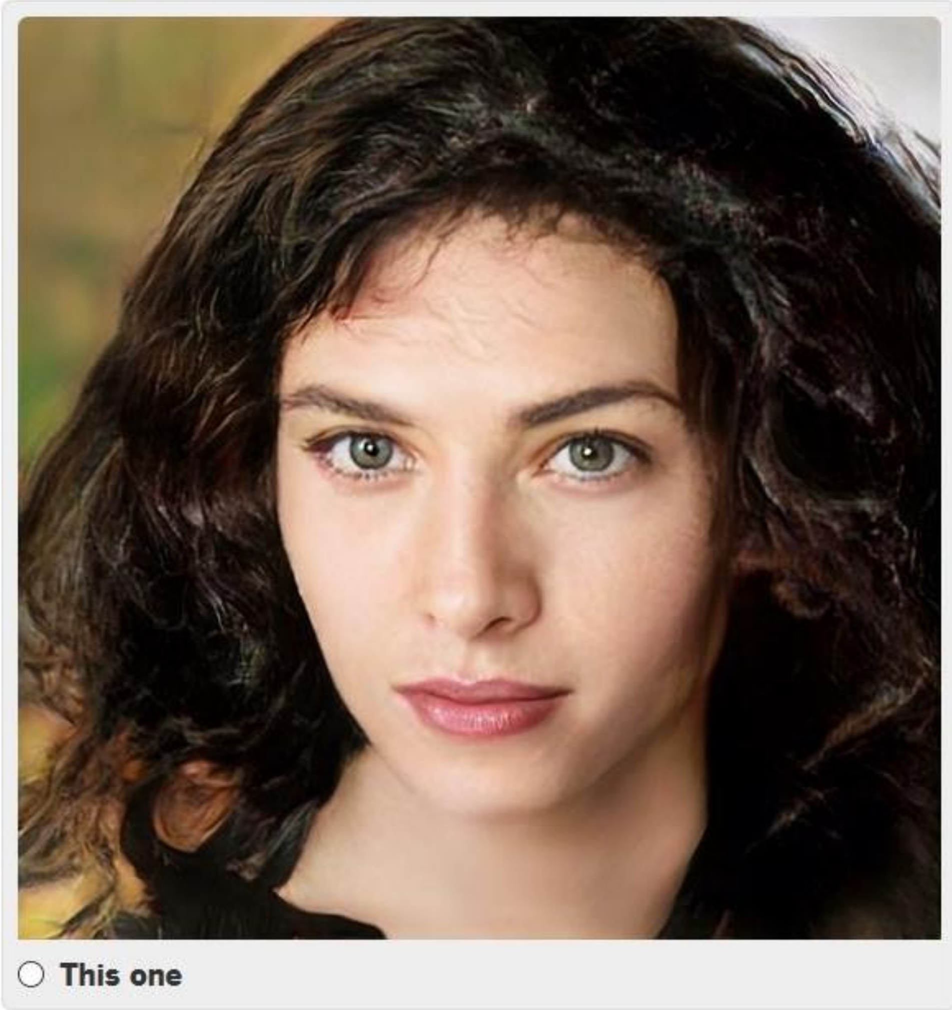

4.8M training games

Strengths & Weaknesses

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worse with **more** data

worse with **less** data

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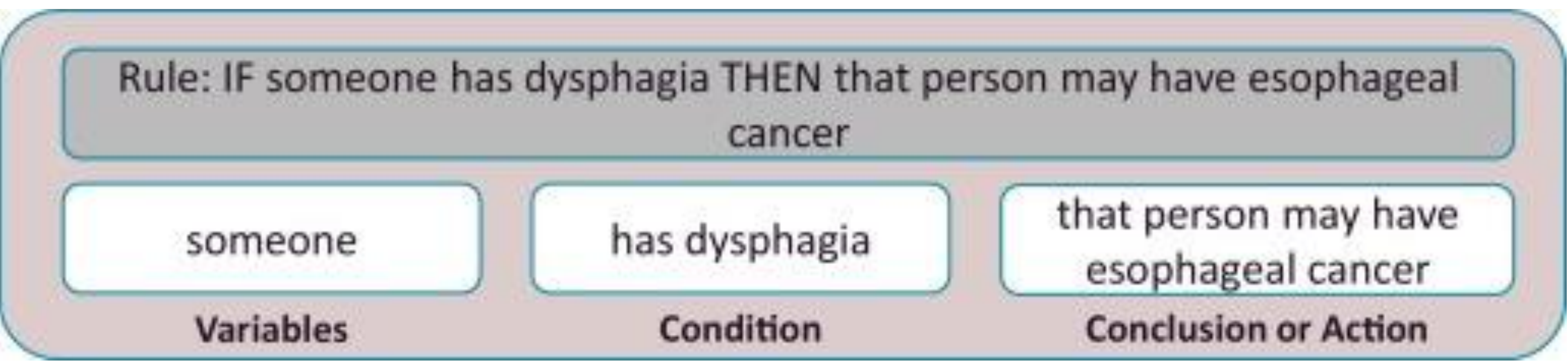

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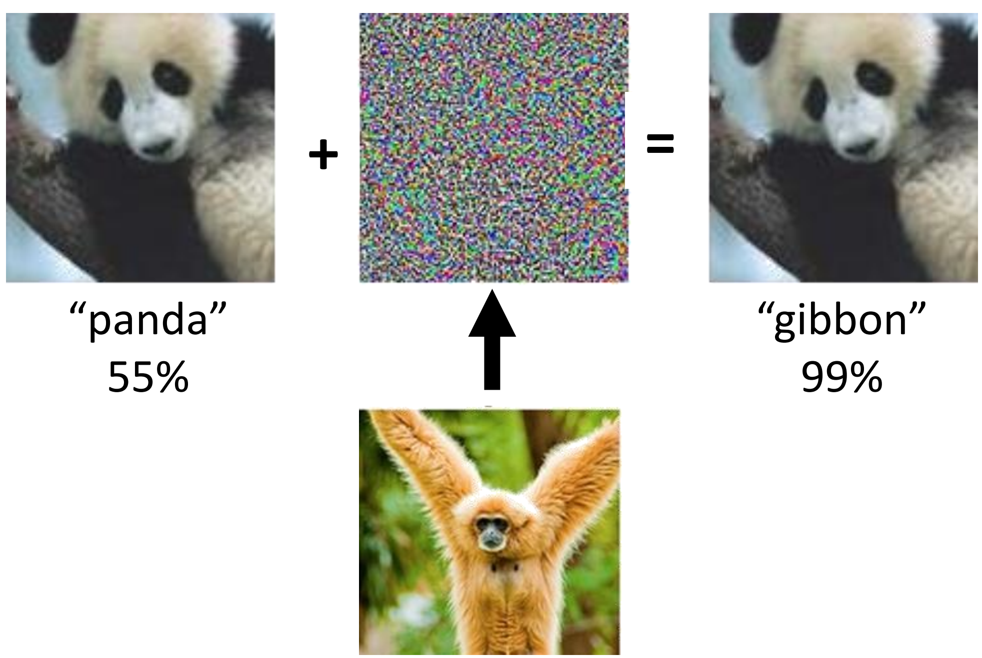

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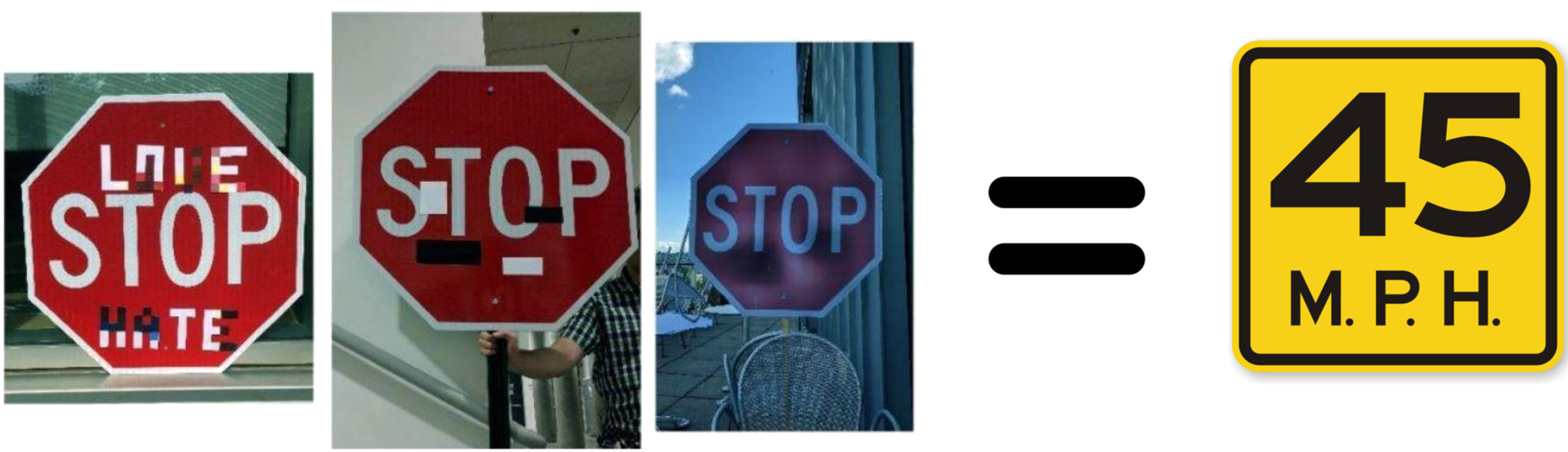

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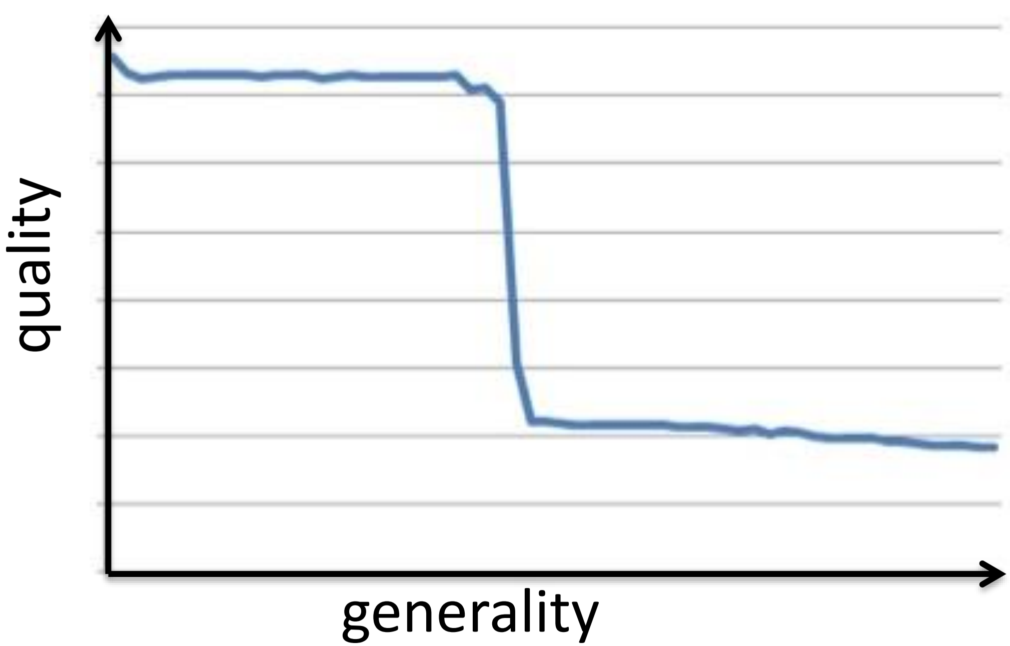

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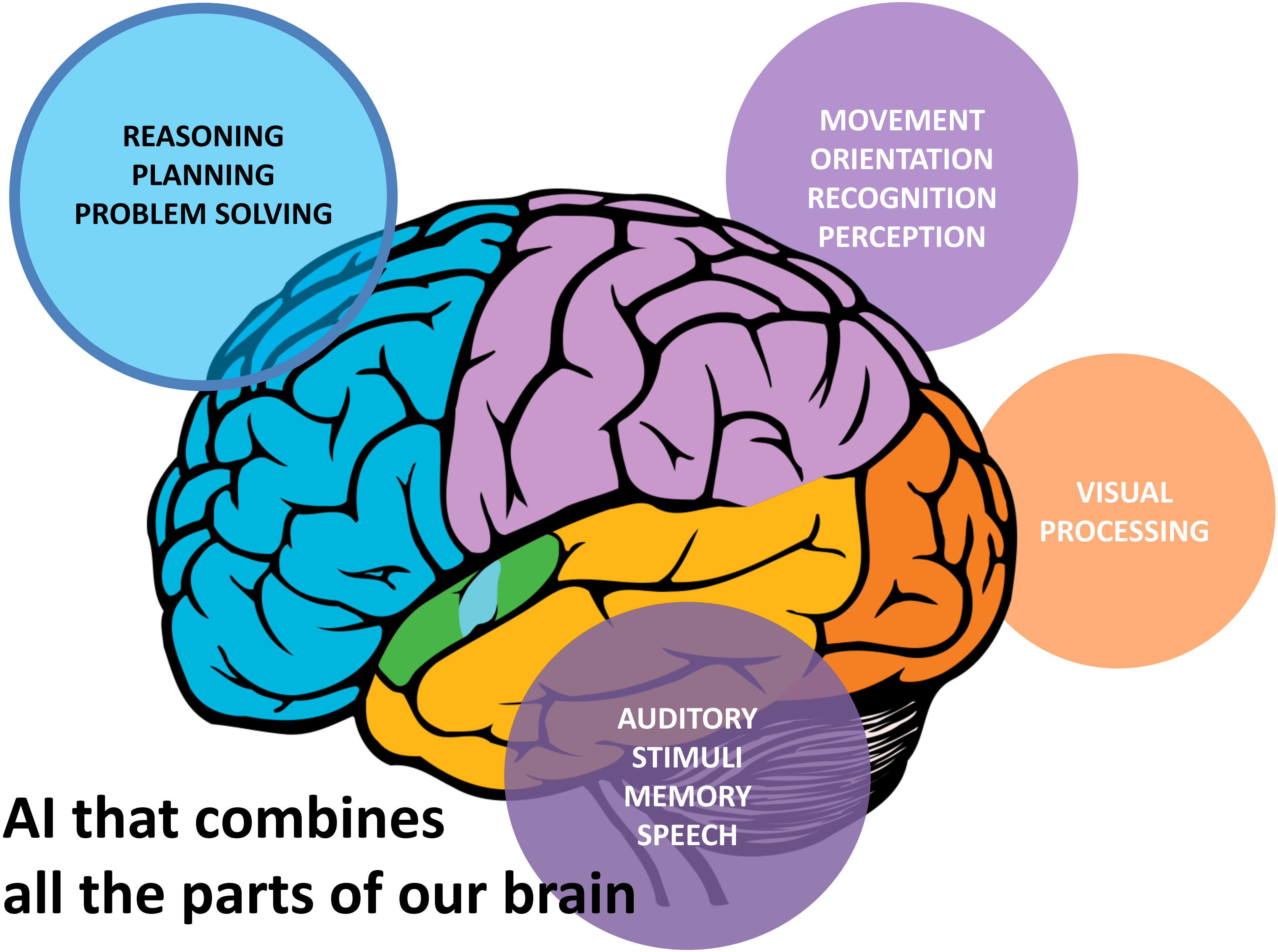
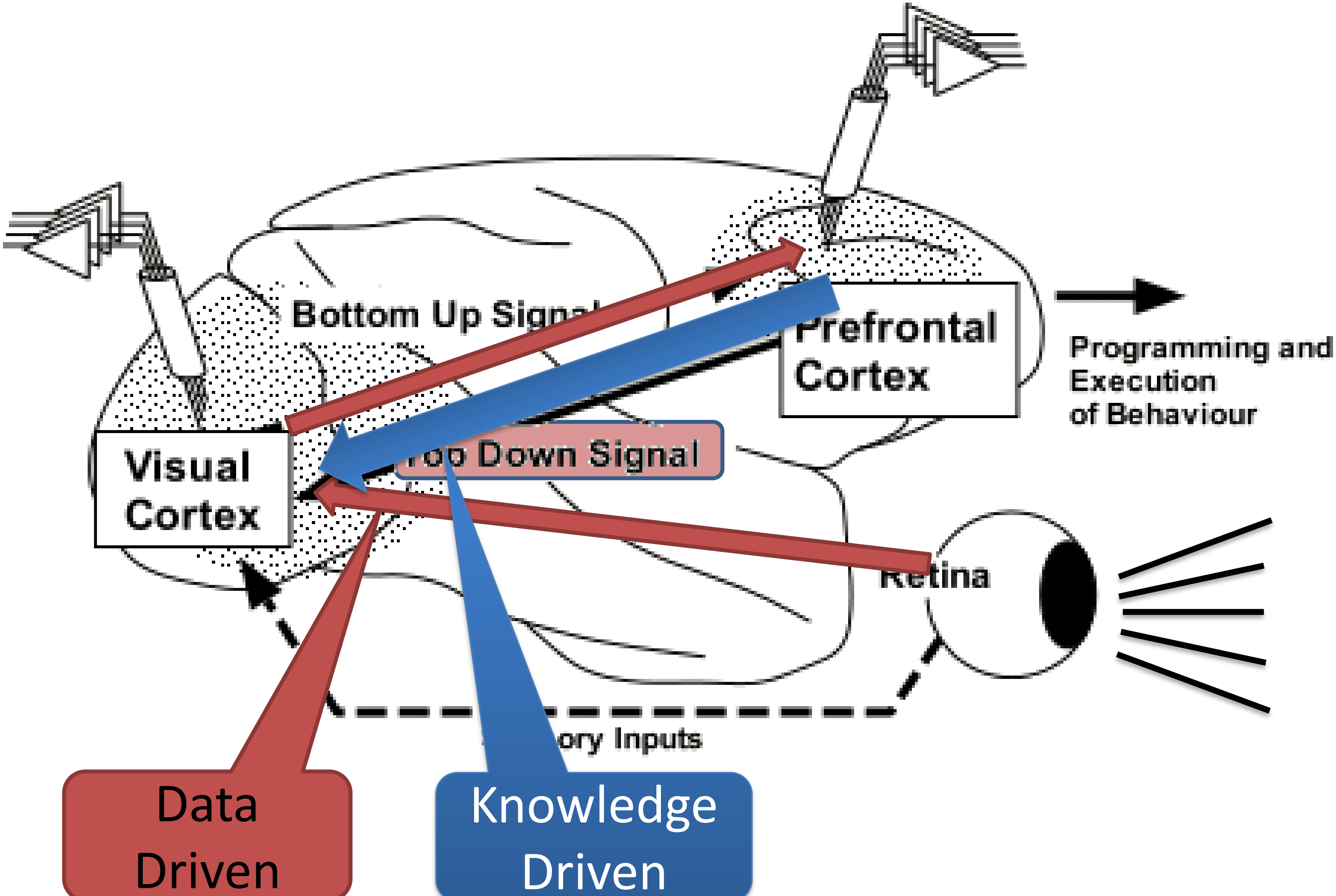


AI that combines all the parts of our brain



Bottom Up
Prefrontal Cortex
Programming and Execution of Behaviour
Visual Cortex
Down Signal
Inputs
Data Driven
Knowledge Driven

Wanted:
AI that combines all the parts of our brain

Learning (Fast)	Reasoning (Slow)
Better with more data	Worse with more data
Needs lots of data	Already works with very little data
Not explainable	Explainable
Robust against noise	Brittle with noise

Handbook of Knowledge Representation (1000 pages, ToC alone is 14 pages)

- propositional logic & satisfiability solvers
- first order logic & resolution
- description logic
- constraint (logic) programming
- nonmonotonic reasoning
- belief revision
- qualitative reasoning
- model-based diagnosis
- bayesian networks
- temporal logic
- spatial reasoning
- epistemic logic
- deontic logic
- situation calculus
- default logic
- event calculus
- **knowledge graphs**
- ……